LoppisFynd

Comprehensive review + improved roadmap + UI/UX asset integration

Focus: remove first-run model download, ship cloud-first AI, keep offline-first core.

2026-02-20

1) Executive summary

This document reviews the current LoppisFynd codebase and the provided UI/UX research pack, then proposes an updated roadmap that keeps the Nature Distilled visual direction while removing the first-run on-device model download as a blocker.

What is already working well

• Offline-first architecture (Drift DAOs as source of truth) with clear service boundaries (AI / market / sync).

• Good delivery hygiene: CI runs format/analyze/custom_lint/tests and builds release bundles for staging + prod.

• Design tokens exist in code (colors/spacing/radius/blur/shadows/typography) and shared “glass” primitives are reusable.

• Custom lints enforce “no ad-hoc design constants” and “no raw UI strings”, which prevents UI drift over time.

Top risks / blockers to address next

• First-run model download (Gemma) is still wired into onboarding + dashboard; this conflicts with the “download-free first run” goal.

• Cloud AI needs a secure proxy (do not call Gemini directly from the client) plus quotas, auth, and a kill-switch.

• App icon is still the default Flutter mark; branding needs a production-ready icon + simplified logo mark.

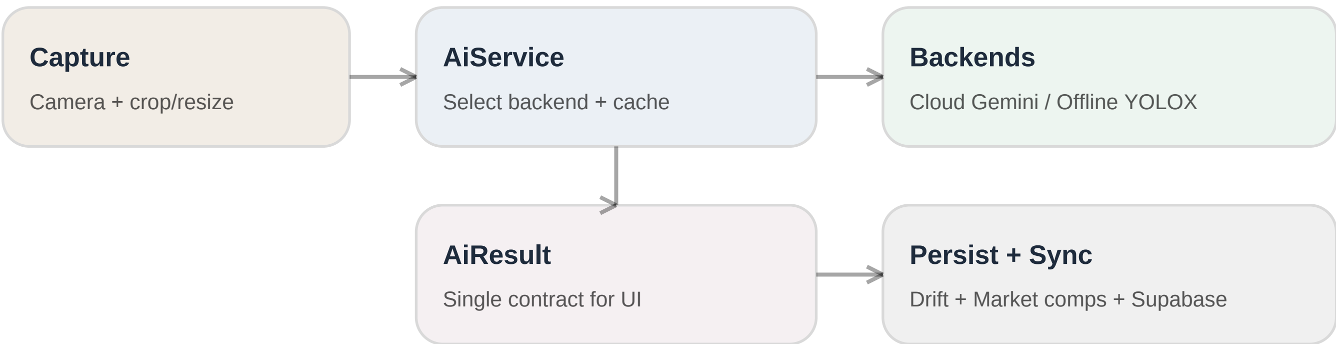
• Dark mode is not implemented yet (only light + high-contrast); tokens need light/dark ramps to scale UI work.

2) Codebase review (repo)

Repo snapshot: 140 Dart files under lib/, 40 Dart tests, 38 planning docs, plus Supabase edge functions and migrations.

Architecture (high-level)

The app is feature-first with a shared core/ platform layer. Drift provides offline persistence; Riverpod is used for DI and reactive state; services handle AI inference, market price comps, and optional Supabase cloud sync. This is a strong base for a cloud-first AI pivot because you can swap implementations behind a single service entrypoint.



AI implementation (current vs target)

Today, AI inference is isolate-based and uses flutter_gemma with a downloadable model managed by ModelInstallController/ModelManager. To meet the adoption goal, the recommended target is: Cloud-first identification (via a secure proxy) + a small optional offline fallback (e.g., YOLOX) - and no mandatory multi-GB downloads on first run.

Dimension	Current (Gemma-era)	Target (Roadmap v3)
First run	User sees consent + can be blocked by model download	No downloads. Cloud identify works immediately; offline is optional.
Privacy & keys	No cloud calls, but large local footprint	Gemini behind secure proxy (no client key exposure), crop-only uploads.
Failure mode	Disk space / slow download / install issues	Network timeouts degrade gracefully: retry or offline quick ID.
Cost/perf	Device storage + local compute	Cloud costs controlled via quotas + image size caps; cache results in Drift.

Concrete refactors (low-risk, high impact)

• Introduce an AiBackend interface and AiService router (cloud/offline/auto) so UI code calls one place.

• Define a single AiResult contract (JSON schema) used by all backends and persisted in Drift.

• Replace Gemma consent/download surfaces with an “AI status” card (Cloud / Offline / Disabled) + settings toggles.

• Reuse the existing Supabase edge-function pattern (used for Tradera proxy) for an /ai/identify proxy.

3) Improved roadmap (2026–2027)

This version merges the repo's Nature Distilled UI work with the updated "download-free first run" AI direction. Milestones are designed to be shippable, measurable, and reversible (feature flags).

When	Milestone	Shippable outcome (Definition of Done)
Q1 2026	M1 — Cloud-first AI, download-free first run	Fresh install is usable immediately. Identify uses cloud by default. No model download required. Kill-switch + metrics in place.
Q2 2026	M2 — Lightweight offline fallback	Offline "quick ID" works in airplane mode using a small model (<10MB) and returns bounding boxes in AiResult.
Q2–Q3 2026	M3 — Dependency + platform modernization	Flutter + core packages upgraded incrementally; builds green; no deprecated APIs; CI stable.
Q3 2026	M4 — Dark mode	Token-driven dark theme across all screens; contrast checks; goldens updated.
Q3–Q4 2026	M5 — Responsive layouts	Breakpoints + intentional tablet/foldable layouts for key screens; no overflow.
Q4 2026	M6 — Token formalization	Tokens in a single JSON source of truth (W3C style), exported to Dart and documented.
Q1–Q2 2027	M7 — Future-proofing	Small, measured adoption of 2027 UI/UX trends; performance profiling; decision log.

Critical path (next 3 sprints)

• Sprint 0 (1–2 days): analytics + Sentry breadcrumbs for onboarding + identify; add kill-switch flags.

• Sprint 1: AiResult contract + AiBackend/AiService abstraction; wire UI to use the abstraction (no behavior change yet).

• Sprint 2: Cloud Gemini proxy + app client integration; remove any Gemma auto-download entrypoints; ship settings toggles.

• Sprint 3: integration tests + staged rollout; then start YOLOX offline fallback.

Mapping to the existing repo roadmap

The repo's .planning roadmap is organized as: (1) tokens/primitives, (2) capsule navigation shell, (3) onboarding/auth + model download, (4) core screens + goldens. Phases 1–2 are already aligned with Nature Distilled and should be kept. Phase 3 should be rewritten: replace "Gemma download" with "Cloud AI status + optional offline fallback".

4) UI/UX pack integration (images + research)

The provided UI/UX pack includes token specs (color/typography/spacing), component guidance (glass + textures), and per-screen layout specs. The codebase already has a strong token/primitives baseline; the main work is aligning token ramps (especially dark mode) and adding texture assets in a performance-safe way.

Pack intent	Current code (AppColors)	Recommendation
Warm neutral background + surface	cloudDancer / clay	Keep as v1, but add neutral ramps (neutral0–neutral4) so borders/elevation scale.
Ink ramp for text	deepSapphire + alpha variants	Add ink0/ink1/ink2 tokens and map typography to semantic roles.
Terracotta brand ramp	terracottaClay + copperOak	Add terracotta ramp (tint→pressed) and use for CTA/selected states.
Dark mode	Not implemented	Implement dark palette + glass rules; add golden tests for core components.

Images: what is safe to ship vs “inspiration only”

Ship-ready assets are textures and the provided logo image. Two reference photos are watermarked (ref1.jpg, ref2.jpg) and should NOT be shipped; keep them as moodboard references only.

File	Type	Suggested new name	Use in app (where)
generated_image_d8a4... png	Texture (paper tile)	Assets/images/textures/paper_worn_tile.jpg	Splash/onboarding/empty-state backgrounds (Stack under glass).
furniture_items.png	Texture (linen tile)	Assets/images/textures/linen_tile.jpg	Cards & sheets overlay (10–20% opacity).
vintage_sweaters.png	Texture (wood tile)	Assets/images/textures/wood_tile.jpg	Detail headers, accent strips, subtle overlay.
generated_image_8380... png	Texture (patina coin)	Assets/images/textures/patina_circle.jpg	Icon backplates / accent badges (e.g., profit tag).
generated_image_cc74... png	Hero photo	Assets/images/photos/flea_market_1.jpg	Onboarding hero, marketing, optional dashboard hero (blurred).
generated_image_9014... png	Hero photo	Assets/images/photos/flea_market_2.jpg	Onboarding hero variant; app store screenshots.
loppis_background.png	Hero texture	Assets/images/photos/linen_plate.jpg	Splash/empty states; background behind AtmosphericBackground.
logo.jpg / Assets/unnamed.jpg	Logo concept	Assets/brand/logo_lockup.jpg	About screen, onboarding, marketing. (Do not use as app icon.)
ref1.jpg, ref2.jpg	Watermarked refs	(do not ship)	Moodboard only. Replace with licensed/owned imagery.

Ship-ready thumbnails (from the pack)

Paper tile
generated_image_d8a4ceab-d797-4adf-9f76-21bcb62adc8a.png

Linen tile
furniture_items.png

Wood tile
vintage_sweaters.png

Patina accent
generated_image_8380e2e4-7692-41a6-961f-fe0aa4e6e693.png

5) Where to wire UI assets in the Flutter code

Below is a pragmatic “what to change where” list for adding the pack’s textures and replacing the current logo-motif usage.

Goal	File(s) to edit	Change
Add texture overlay behind the existing AtmosphericBackground	lib/shared/widgets/atmospheric_background.dart	Add an optional DecorationImage overlay (low opacity) before the painter track. Gate behind a setting/flag.
Use paper/linen textures for onboarding & empty states	lib/features/onboarding/onboarding_screen.dart; lib/shared/widgets/empty_state.dart	Wrap layouts in a Stack: texture image -> tinted fill -> glass content.
Replace LogoMotifOverlay image with a proper “mark-only” asset	lib/shared/widgets/logo_motif_overlay.dart	Swap AssetImage from Assets/unnamed.jpg to Assets/brand/mark.png (a simplified icon).
Remove Gemma download UI and replace with “AI Status: Cloud/Offline”	lib/features/dashboard/dashboard_screen.dart; lib/features/model_manager/*; lib/features/onboarding/*	Delete or repurpose ModelPreflightCard. Add a status card that reflects AiService backend state.
Add AI settings toggles and enforce them	lib/features/settings/settings_screen.dart + app settings DAO	Implement: “Cloud AI”, “Send crops only”, “Fetch sold comps”, “Offline quick ID”.

Recommended repo asset structure

Move from single-file assets to folder-based assets so you can add more textures without touching pubspec each time:

```
Assets/
brand/
logo_lockup.jpg
mark.png
images/
photos/
flea_market_1.jpg
flea_market_2.jpg
textures/
paper_worn_tile.jpg
linen_tile.jpg
wood_tile.jpg
patina_circle.jpg
```

6) Logo & app icon recommendations

Current situation: the app uses a photo-based circular logo (Assets/unnamed.jpg) for motif overlays, and the actual app icon is still the default Flutter mark. Recommendation: keep the current lockup for marketing/about screens, but create a simplified “mark-only” icon for the app launcher and for subtle motif tiling.

Current lockup (good for About / marketing)

Use cases: About screen, onboarding header, social sharing card. Avoid as launcher icon (too much detail at 48–96px).

Suggested launcher icon (concept)

A launcher icon should read at small sizes: one shape, one idea. Below is a simple vector concept aligned with the palette: terracotta ring + magnifier dot + “L” path. (This is a direction, not a final brand asset.)

Production checklist for branding

- Create mark-only SVG (and export PNGs) for launcher icons (Android mipmaps + iOS app icon set).
- Create monochrome mark for in-app motif overlay (works with tint + alpha).
- Keep the full lockup image for About/marketing; avoid tiling a photo logo in UI for performance.
- Verify license provenance for any third-party photos; remove watermarked refs from shipping assets.

Appendix: Key references (files)

UI/UX research pack (provided)

Key system specs: color_palette_tokens.md, typography_tokens.md, spacing_elevation_tokens.md, glassmorphism_components.md, textures.md. Per-screen specs exist for login, onboarding, dashboard, scanner, history, drafts, item detail, hauls, settings, privacy, sync status, and account deletion.

File	What it covers
account_deletion_screen.md	Per-screen layout + components + states
auth_gate_screen.md	Per-screen layout + components + states
color_palette_tokens.md	Design tokens
color_trends_2026.md	Reference
color_trends_behance_2026.md	Reference
dashboard_screen.md	Per-screen layout + components + states
dashboard_screen_research_2026.md	Trend scan / rationale
deep_link_gate_screen.md	Per-screen layout + components + states
draft_editor_screen.md	Per-screen layout + components + states
drafts_screen.md	Per-screen layout + components + states
glassmorphism_components.md	Glass component rules
glassmorphism_research_2026.md	Trend scan / rationale
haul_screen.md	Per-screen layout + components + states
haul_summary_screen.md	Per-screen layout + components + states
history_screen.md	Per-screen layout + components + states
item_cards.md	Reference
item_cards_research_2026.md	Trend scan / rationale
item_detail_screen.md	Per-screen layout + components + states
login_screen.md	Per-screen layout + components + states
model_manager_screen.md	Per-screen layout + components + states
… (+15 more)	

Repo files to touch for the cloud-first AI pivot (starting points)

Area	Files
AI model download + consent	lib/services/ai/model_install_controller.dart; lib/features/onboarding/onboarding_screen.dart
Gemma backend	lib/services/ai/inference/flutter_gemma_backend.dart; lib/services/ai/inference/inference_isolate_service.dart

Dashboard model preflight card	lib/features/dashboard/dashboard_screen.dart
Settings toggles	lib/features/settings/settings_screen.dart; lib/core/database/daos/app_settings_dao.dart
Edge function pattern	supabase/functions/tradera-proxy (use as template for /ai/identify)
Feature flags	lib/core/config/feature_flags.dart

Note: this PDF intentionally focuses on “what to ship next” rather than repeating every planning document. The codebase already contains a lot of good planning detail; the main delta is replacing the Gemma download path with a cloud-first AI path while keeping the offline-first UX intact.